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April 19, 2026

The Editorial Board

Journal of Computational Neuroscience

Springer Nature

Dear Editors,

I am pleased to submit an original research manuscript entitled **Longitudinal Boundary Sharpness Coefficient Slopes Predict Time to Alzheimer Disease Conversion in Mild Cognitive Impairment: A Survival Analysis Using the ADNI Cohort** for consideration in the *Journal of Computational Neuroscience*.

Predicting which individuals with mild cognitive impairment will progress to Alzheimer disease remains one of the most clinically consequential unsolved problems in computational neuroscience. Most existing approaches rely on a single baseline scan, yet the structural brain changes that signal impending conversion unfold over months and years. This manuscript addresses that gap by introducing longitudinal slope features derived from the Boundary Sharpness Coefficient, a voxel-level measure of gray-white matter interface integrity computed from standard T1-weighted MRI. Across 1,824 scans from 450 ADNI participants, annualized rates of boundary degradation modeled with a Random Survival Forest achieved a concordance index of 0.63, representing a 163 percent improvement over a baseline parametric model. The method requires no PET imaging, no cerebrospinal fluid collection, and no sequence beyond the structural MRI already obtained in routine clinical evaluations.

This work sits at the intersection of computational neuroimaging, survival analysis, and machine learning applied to neurodegeneration, areas that fall squarely within the scope of the *Journal of Computational Neuroscience*. The manuscript will be of direct interest to readers working on quantitative brain imaging biomarkers, longitudinal modeling of neural tissue, and data-driven approaches to clinical risk stratification. The proposed framework is reproducible, interpretable, and designed with real-world deployment constraints in mind.

This manuscript is original, has not been published previously, and is not currently under consideration by any other journal. I have no competing interests to disclose. As the sole author, I have approved the final version of the manuscript and take full responsibility for its content.

Thank you for your time and consideration. I look forward to your response.

Sincerely,

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